# PRACTICAL NO. 8

A system has RAM which can store four pages simultaneously to satisfy page requests. Write a Program to calculate percentage page hit and page fault if the system uses Least recently used page replacement technique for the page references entered by the user i.e. 1, 2, 3, 4, 1, 2, 5, 1, 2, 3, 4, 5 (No of Frames=3).

**CODE:**

```
MINGW64:/d/OS_CM24112/OSLab_CM24112/Practiacl_8
GNU nano 8.5                          lru.c                          Modified
        if(!found) {
            int lru_index = 0;

            for(j = 1; j < frames; j++) {
                if(time[j] < time[lru_index]) {
                    lru_index = j;
                }
            }

            memory[lru_index] = pages[i];
            counter++;
            time[lru_index] = counter;
            fault++;
        }
    }

    printf("Total Hits = %d\n", hit);
    printf("Total Faults = %d\n", fault);

    float hit_ratio = (float)hit / n * 100;
    float fault_ratio = (float)fault / n * 100;

    printf("Hit Percentage = %.2f%%\n", hit_ratio);
    printf("Fault Percentage = %.2f%%\n", fault_ratio);

    return 0;
}
```

MINGW64:/d/OS_CM24112/OSLab_CM24112/Practiacl_8

```
  GNU nano 8.5                          lru.c                           Modified
#include <stdio.h>

int main() {
    int frames = 3;
    int pages[] = {1,2,3,4,1,2,5,1,2,3,4,5};
    int n = 12;

    int memory[3];
    int time[3];
    int i, j;
    int hit = 0, fault = 0;
    int counter = 0;

    // Initialize memory
    for(i = 0; i < frames; i++) {
        memory[i] = -1;
        time[i] = 0;
    }

    for(i = 0; i < n; i++) {
        int found = 0;

        // Check HIT
        for(j = 0; j < frames; j++) {
            if(memory[j] == pages[i]) {
                hit++;
                counter++;
                time[j] = counter;
                found = 1;
                break;
            }
        }
```

**OUTPUT:**

MINGW64:/d/OS_CM24112/OSLab_CM24112/Practiacl_8

```
Shars@Shars-PC MINGW64 /d/OS_CM24112/OSLab_CM24112/Practiacl_8 (main)
$ nano lru.c

Shars@Shars-PC MINGW64 /d/OS_CM24112/OSLab_CM24112/Practiacl_8 (main)
$ gcc lru.c -o lru
./lru
Total Hits = 2
Total Faults = 10
Hit Percentage = 16.67%
Fault Percentage = 83.33%
```